

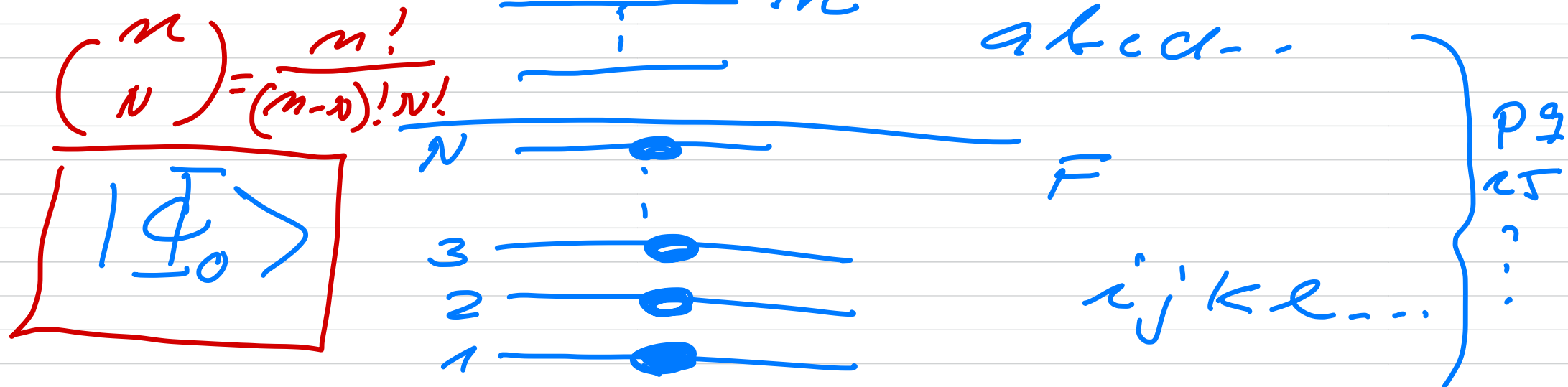
FYS4480/9480 september 11



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$|\Phi_0\rangle$ ($|c\rangle$) ansatz

for ground state, leads
to the new vacuum, or
just reference state



Wick's - theorem

$$xy = \langle 0|xy|0\rangle + N[xy]$$

$$= \overline{xy} + N[xy]$$

$$a_p^\dagger a_q + a_q a_p^\dagger = \delta_{pq}$$

$$a_p|0\rangle = 0 \quad a_p|p\rangle = |0\rangle$$

$$a_p^\dagger|0\rangle = |p\rangle \quad a_p^\dagger|p\rangle = 0$$

$$a_p a_q^\dagger = \overline{a_p a_q^\dagger} + N[a_p a_q^\dagger]$$

$$a_p^\dagger a_q = \overline{a_p^\dagger a_q} + N[a_p^\dagger a_q]$$

$$\langle 0 | a_p^\dagger a_q | 0 \rangle \rightarrow \langle c | a_p^\dagger a_q | c \rangle$$

$\stackrel{!}{=} 0$

$$a_q | c \rangle = a_q | \Phi_c \rangle$$

$$q \in \{1, \dots, N\}$$

$$a_q a_1^\dagger a_2^\dagger \dots a_q^\dagger \dots a_N^\dagger | 0 \rangle$$

$$= (-1)^{q-1} | \Phi_q \rangle$$

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$N-1$  SD

$$a_i |\Phi_0\rangle \neq 0 \quad \text{if } i \in \{\Phi_0\}$$

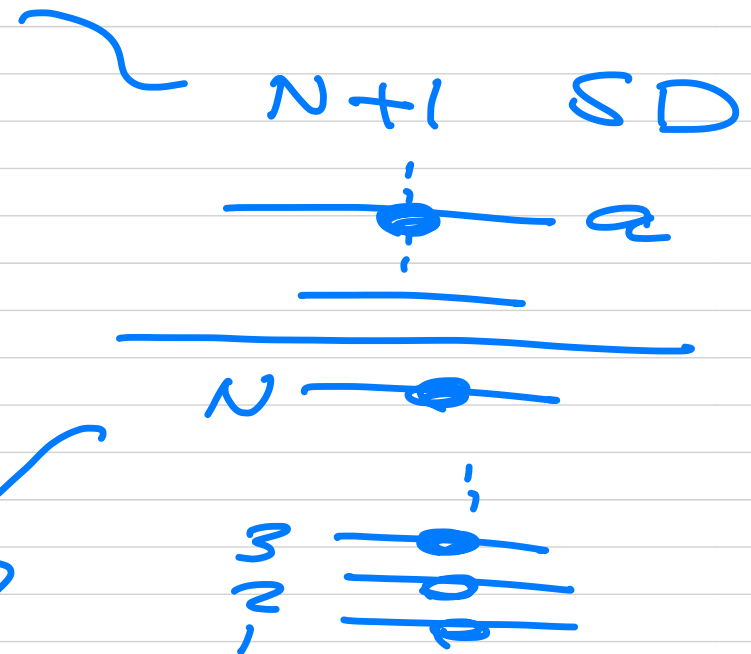
$$a_i^+ |\Phi_0\rangle = 0 \quad i, j, k, l, \dots \leq F$$

$$a_a^+ |\Phi_0\rangle = (-1)^N |\Phi^a\rangle a_b c d \dots > F$$

$$a \notin \{\Phi_0\}$$

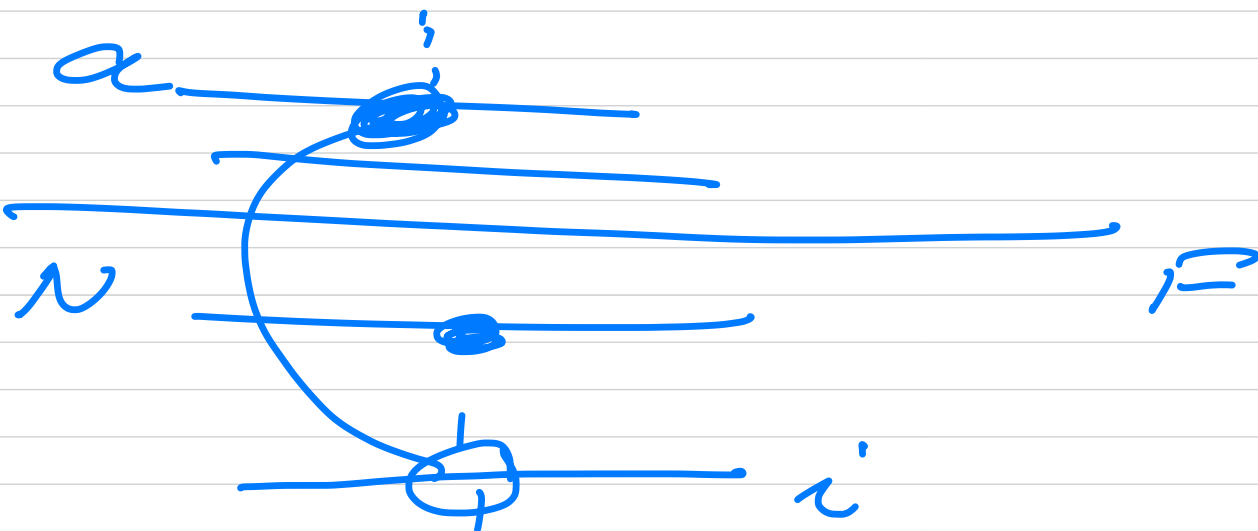
$$a_a |\Phi_0\rangle = 0$$

$$|\Phi^a\rangle$$

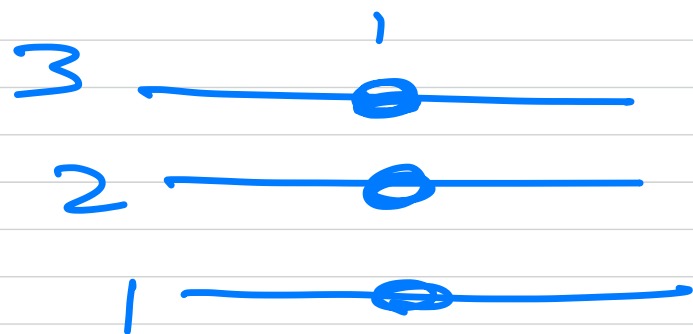


1ph excitation

1 1 1 1 1 n

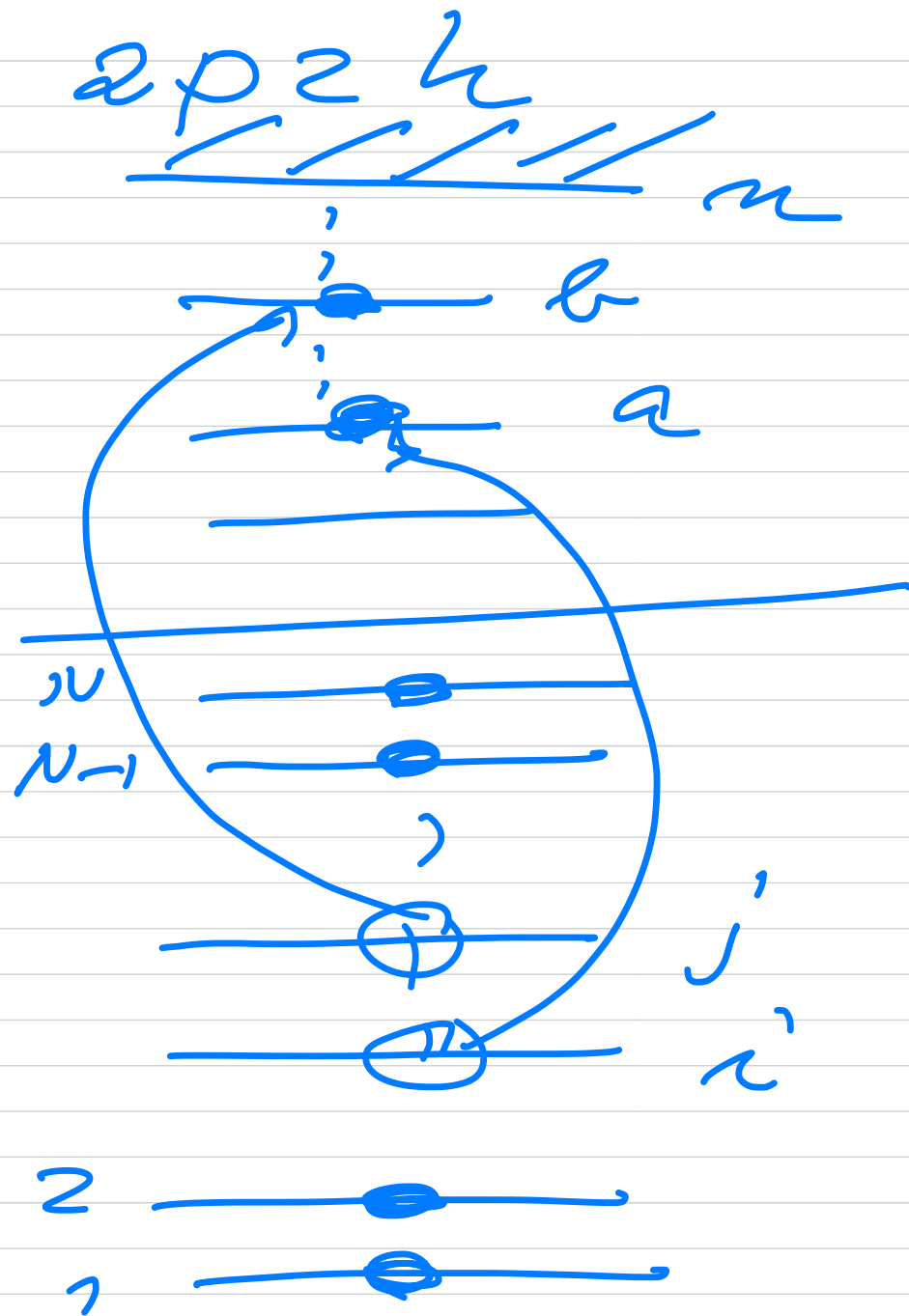


$$a_a^\dagger a_{i'} |\Phi_0\rangle$$



$$= |\Phi_n^a\rangle$$

N-particle  
SD



$$|\Phi_{ij}^{ab}\rangle$$

$$a_a^+ a_b^+ a_j a_i |\Phi_0\rangle$$

$N$ -particle  
SD

with  $m$ -single-particle  
states and  $N$ -particles  
we can make

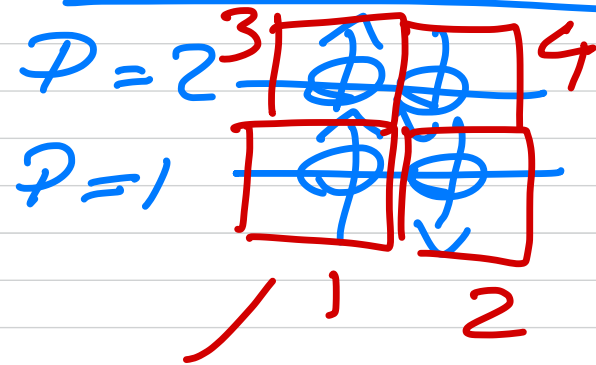
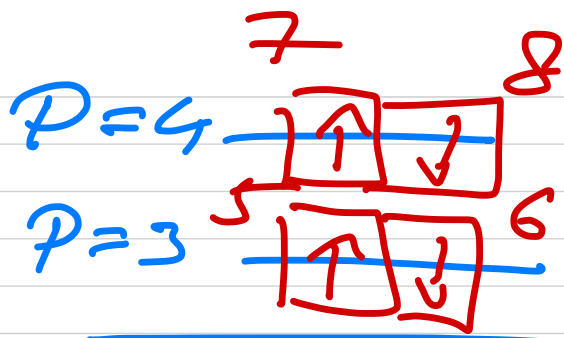
$$\binom{m}{N} = \frac{m!}{(m-N)! N!}$$

configurations

Example: pairing  $\uparrow\downarrow$

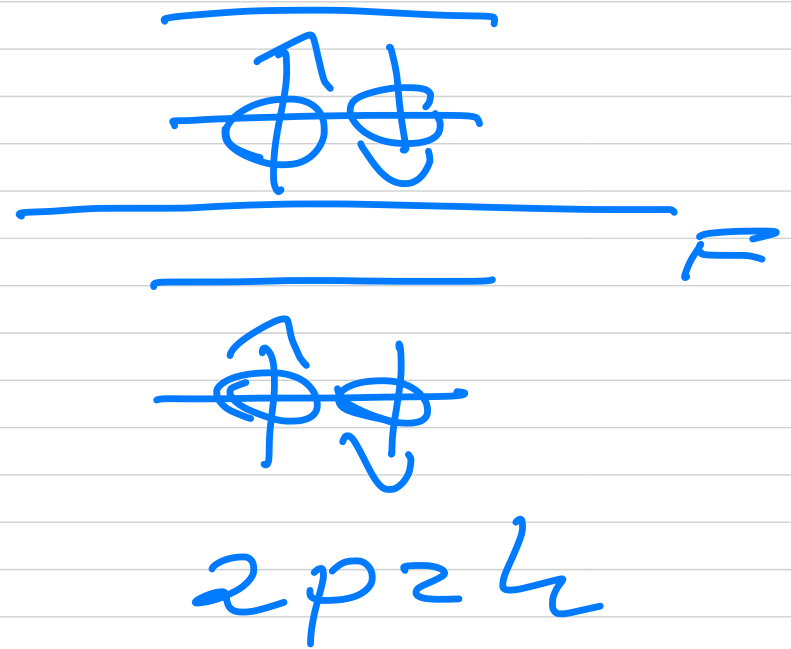
$$m = 8 \quad N = 4$$

$$\binom{8}{4} = 70$$

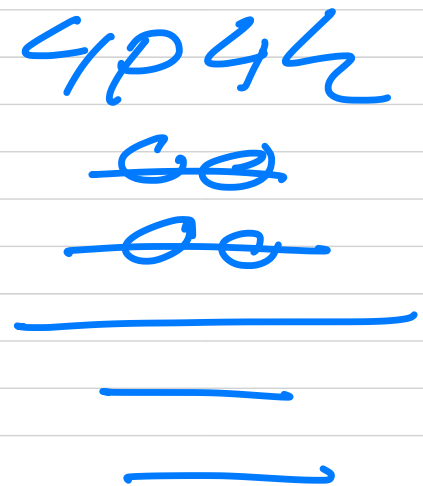
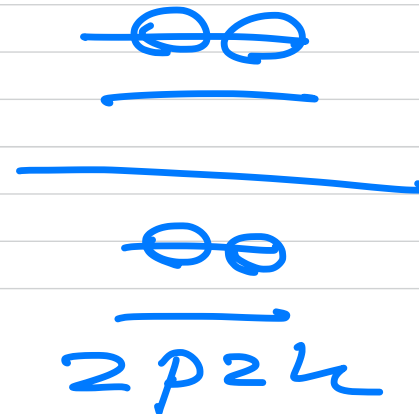
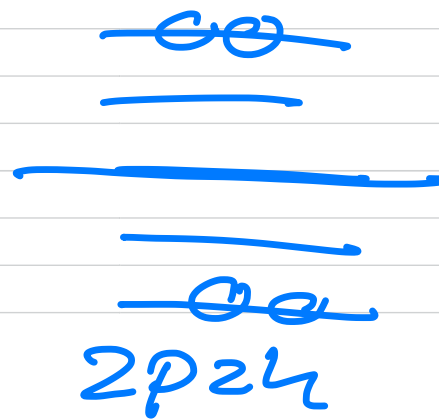
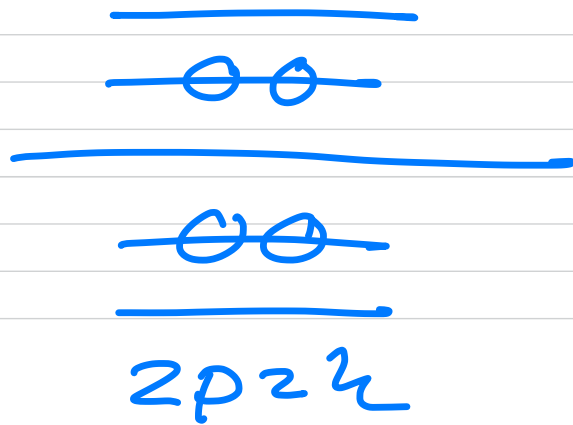


$P=1, \uparrow$

$|\Phi_0\rangle$



$$\binom{4}{2} = 6$$



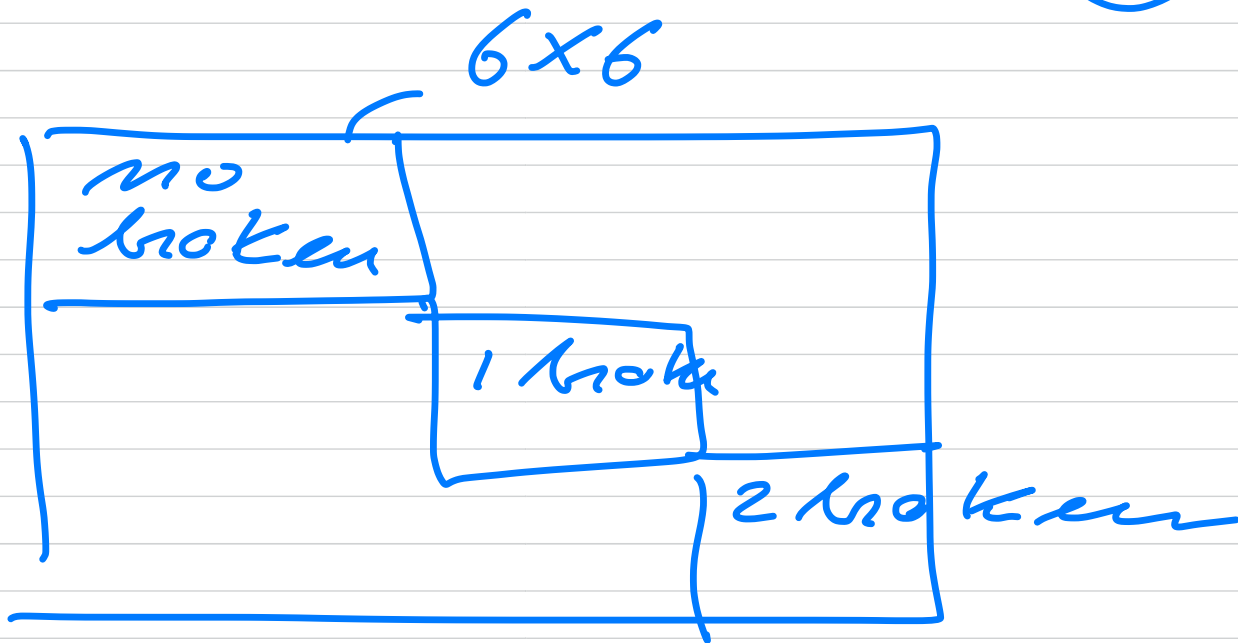
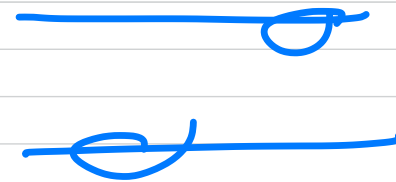
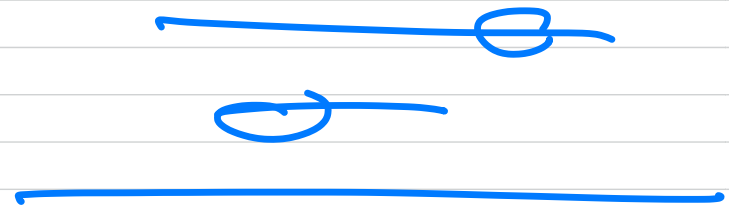
$$|\underline{\psi}_0\rangle = \sum_i c_{0i} |\underline{\Phi}_i\rangle$$

$$c_{0i} = \langle \underline{\psi}_0 | \underline{\Phi}_i \rangle$$

$$|\underline{\psi}_0\rangle = \sum_{PH} C_H^P |\underline{\Phi}_H^P\rangle$$

$$= C_0^0 |\underline{\Phi}_0\rangle + \sum_{a_i} C_a^a |\underline{\Phi}_a^a\rangle$$

$$+ \sum_{\substack{a,b \\ i,j}} C_{ij}^{ab} |\underline{\Phi}_{ij}^{ab}\rangle + \dots \quad \sum_{N \neq N+1} C_{HN}^{PN} |\underline{\Phi}_{HN}^{PN}\rangle$$



$$a_p a_q^\dagger = \overline{a_p a_q^\dagger} + N[a_p a_q^\dagger]$$

$$a_p^\dagger a_q = \underbrace{\overline{a_p^\dagger a_q}}_{N[a_p^\dagger a_q]} + N[a_p^\dagger a_q]$$

$$= \{a_p^\dagger a_q\}$$

normal-  
ordering  
wrt  $|c\rangle$   
( $|\Phi_c\rangle$ )

$$\hat{H}_0 = \sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q$$

$$= \sum_{pq} \langle p | \hat{h}_0 | q \rangle \{ a_p^\dagger a_q \}$$

$$+ \sum_i \langle i | \hat{h}_0 | i \rangle$$

$$a_p^\dagger a_q = \underbrace{a_p^\dagger a_q}_{\delta_{pq} \text{ if } p, q \in \{1, 2\}} + \{ a_p^\dagger a_q \}$$

$$\delta_{pq} \text{ if } p, q \in \{1, 2\}$$

$$\delta_{pq} \in i$$

$$\underbrace{a_p^\dagger a_q}_{\delta_{pq}} = 0$$

$$a_p^\dagger a_q^\dagger a_s a_r = N [a_p^\dagger a_q^\dagger a_s a_r]$$

$$+ N [a_p^\dagger \overbrace{a_q^\dagger a_s}^+ a_r]$$

$$+ N [\overbrace{a_p^\dagger a_q^\dagger}^+ a_s a_r]$$

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$$+ N [a_p^\dagger \overbrace{a_q^\dagger a_s a_r}^+]$$

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$$+ N [\overbrace{a_p^\dagger a_q^\dagger a_s a_r}^+]$$

$$\delta_{qs} N [a_p^\dagger a_r]$$

$qs \in \Lambda'$

$$- \delta_{ps} N [\overbrace{a_q^\dagger a_r}^+]$$

$ps \in \Lambda'$

$$+ \delta_{pr} N [a_q^\dagger a_s]$$

$$- \delta_{qr} N [a_p^\dagger a_s]$$

$qr \in \Lambda'$

$$+ \delta_{pr} \delta_{qs}$$

$pr \in \Lambda' \quad qs \in \Lambda'$

$$- \delta_{ps} \delta_{qr}$$

$ps \in \Lambda' \quad qr \in \Lambda'$

$$H_I = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$$

$$= \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$+ \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \left[ \delta_{rs \in i} \{ a_p^\dagger a_r \} \right.$$

$$- \delta_{rs \in i} \{ a_p^\dagger a_s \} - \delta_{ps \in i} \{ a_q^\dagger a_r \}$$

$$+ \delta_{pr \in i} \{ a_q^\dagger a_s \} + \delta_{pr \in i} \delta_{rs \in i} - \delta_{ps \in i} \delta_{rs \in i} \left. \right]$$

$$\frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \delta_{rs \in i} \{a_p^\dagger a_q\}$$

$$= \frac{1}{4} \sum_{p r i} \langle p i | v | r i \rangle_{AS} \{a_p^\dagger a_r\}$$

next

$$- \frac{1}{4} \sum_{p s i} \langle p i | v | i s \rangle_{AS} \{a_p^\dagger a_s\}$$

$$= \langle p i | v | s i \rangle$$

$$i \longleftrightarrow s$$

$$\hat{H}_I = \frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS} \\ \times \{ a_p^\dagger a_q^\dagger a_s a_r \}$$

$$+ \sum_{pq} \langle p | v | q \rangle_{AS} \{ a_p^\dagger a_q \}$$

$$+ \frac{1}{2} \sum_{i'j'} \langle i'j' | v | i'j' \rangle_{AS}$$

$$\hat{H} = \hat{H}_0 + \hat{H}_I = E_{Ref} + \hat{F}_N + \hat{V}_N$$

$$E_{\text{ref}} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle$$

$$= \sum_i \langle i | \hat{h}_0 | i \rangle + \frac{1}{2} \sum_{i,j} \langle i,j | v | i,j \rangle_{\text{As}}$$

$$\hat{F}_N = \sum_{pq} \langle p | \hat{h}_0 | q \rangle \{ a_p^\dagger a_q \}$$

$$+ \sum_{pq, i} \langle p i | v | q i \rangle_{\text{As}} \{ a_p^\dagger a_q \}$$

$$= \sum_{pq} \langle p | f | q \rangle \{ a_p^\dagger a_q \}$$

$$f_{pq} = \langle p | f | q \rangle = \langle p | h_0 | q \rangle + \sum_i \langle p i | v | q i \rangle_{\text{As}}$$

$$\hat{V}_N = \frac{1}{4} \sum_{pq, r} \langle pq | v | rs \rangle \{ a_p^\dagger a_q^\dagger a_r a_s \}$$

$$\langle \Phi_c^q | H | \Phi_0 \rangle \quad (\text{condensation})$$

$$(i) \quad E_{ref} \cdot \langle \Phi_c^q | \Phi_c \rangle = 0 \quad 1445$$

$$(ii) \quad \sum_{pq} f_{pq} \langle \Phi_c | a_p^\dagger a_q \overbrace{a_p^\dagger a_q}^{\delta_{pa}} | \Phi_0 \rangle$$

$$= f_{ai} = \langle a | f | i \rangle$$

$$(iii) \quad \langle \Phi_a | \hat{V}_N | \Phi_0 \rangle$$

$$= \frac{1}{4} \sum_{pqr, r's} \langle \Phi_0 | \overbrace{\{a_c^\dagger a_c\} \{a_p^\dagger a_q^\dagger a_r^\dagger a_s\}}^{\text{}} | \Phi_0 \rangle \\ \times \langle pqr | r's | r's \rangle_{A_5}$$

$$= 0 \quad \checkmark$$

$$\langle \Phi_a | \hat{H} | \Phi_0 \rangle = \langle a | f | i \rangle \\ = \langle a | \hat{h}_0 | i \rangle + \sum_{j \in F} \langle a | v | ij \rangle_{AF}$$

$$(HF : \langle a | f | i \rangle = 0)$$

$$\langle \Phi_{ij}^{ar} | \hat{H} | \Phi_0 \rangle$$

$$(i) \quad \langle \Phi_{ij}^{ar} | E_{ref} | \Phi_0 \rangle = 0$$

$$(i') \quad \langle \Phi_{ij}^{ar} | \hat{F}_N | \Phi_0 \rangle$$

$$= \sum_{pq} \langle p | f | q \rangle \langle \Phi_0 | \{ \overbrace{a_j^\dagger a_i^\dagger}^+ a_p a_q \} \times \underbrace{\{ a_p^\dagger a_q^\dagger \}}^+ | \Phi_0 \rangle$$

$$= 0$$

$$(iii) \quad \langle \Phi_{ij}^{ar} | \hat{V}_N | \Phi_0 \rangle = \langle ar | v | ij \rangle_{As}$$

## Exercise

$$a) \quad \langle \Phi_{ijk}^{abc} | \hat{H} | \Phi_0 \rangle = 0$$

$$b) \quad \langle \Phi_i^a | \hat{H} | \Phi_j^b \rangle = ? (\neq 0)$$

$$c) \quad \langle \Phi_{ij}^{ab} | \hat{H} | \Phi_k^c \rangle \neq 0$$

$F(u) \subset (\text{configuration}) I(\text{interaction})$

$$\langle \Phi | A | \Phi \rangle$$

$$|\psi_i\rangle = \sum_{pH} C_H^p |\Phi_H^p\rangle$$

$$\sum_{p'H'} \langle \Phi_H^p | \hat{H} - E | \Phi_{H'}^{p'} \rangle C_{H'}^{p'}$$

$$= 0$$

$$\det(\hat{H} - E\mathbb{1}) = 0$$

Opch 1p1L 2p2L 3p3L 4p4L ...

| Opch | Eref                                       | fia | $\langle \alpha_j   \alpha_i \rangle_{AS}$ | 0 | 0 | - | 0       |
|------|--------------------------------------------|-----|--------------------------------------------|---|---|---|---------|
| 1p1L | fai                                        | x   |                                            | x |   | x | 0 - - 0 |
| 2p2L | $\langle \alpha_i   \alpha_j \rangle_{AS}$ | x   |                                            | x |   | x | x 0 - - |
| 3p3L | 0                                          | x   |                                            | x |   | x | x x 0   |
| 4p4L | 0                                          | 0   |                                            | x |   |   |         |
| :    | 0                                          | 0   |                                            | 0 |   |   |         |
| :    |                                            |     |                                            | 0 |   |   |         |
| NpNL | 0                                          | 0   |                                            | 1 |   |   |         |

$$\langle \phi_0 | H | \phi_0 \rangle =$$

$$\langle \Phi_c | H | \Phi_c \rangle = E_{ref}$$

$$H C = E C$$

$$C = \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_n \end{bmatrix}$$